Descriptions of Generalized IS Functions

`gen_IS_fun`

gen_IS_fun implements a two-stage Bayesian Markov Chain Monte Carlo (MCMC) sampling framework for models with not fully observed covarite ζ using Importance Sampling (IS) and Copula-based approximations. The function supports four sampling algorithms specified via the model parameter:

- "plugin": Fixes ζ to its Stage 1 posterior mean (or other point estimate) across all iterations.
- "IIS" (Independent Importance Sampling)
- "AIS" (Adjusted Importance Sampling)
- "AIS_copula" (Copula + Vecchia AIS)

gen_IS_fun is modular and model-agnostic, accepting user-defined likelihood (lik_fun), parameter initialization (init_params), and sampling (sample_params) functions—enabling seamless adaptation between Gaussian, Negative Binomial (Polya-Gamma), and other non-Gaussian response distributions.

Usage

```
gen_IS_fun(y,
           x = NULL,
           sample_params,
           init_params,
           lik_fun = NULL,
           model = c("AIS", "IIS", "plugin", "AIS_copula"),
           zeta_ppd,
           locs = NULL,
           R = 300,
           m_vecchia = 20,
           mcmc_samples = 1000,
           burn_in = 1000)
```

Arguments

- **y**: A numeric vector of length `n` containing outcome data
- **x**: A `n x p` matrix representing known fixed-effect covariate predictors. Defaults to `NULL`
- **sample_params**: A function used to sample model-level parameters. Inputs `y`, `x`, current draw `zeta_sam`, and a named list `params`containing current draw of model-level parameters we are conditioning on.
- **init_params**: A function returning a named list of initial values for model parameters.
- **lik_fun**: A function evaluating an `S x n` log-likelihood matrix of `y` conditioned on Stage 1 partial posterior draws and current parameters. Defaults to `NULL` and is not used by the plug-in method.
- **model**: A character string specifying the sampling algorithm. Accepted options are:
  - "plugin": Fixed spatial latent process, set to Stage 1 posterior means by default.
  - "IIS": Independent Importance Sampling.
  - "AIS": Adjusted Importance Sampling.
  - "AIS_copula": Adjusted Importance Sampling with continuous Empirical CDF transformation, spatial Gaussian Copula, and Vecchia likelihood re-weighting.

- **zeta__ppd**: A numeric matrix of dimension `S × n` containing `S` posterior predictive draws of unobserved variable ζ obtained from Stage 1. For "plugin", may also be a numeric vector of length `n` containing a point estimate of ζ obtained from Stage 1.
- **locs**: A numeric matrix of dimension `n × 2` specifying spatial coordinates (x_1, x_2) for each location. Used only when model = "AIS_copula".
- **R**: An integer defining the number of discrete candidate draws resampled in the second sampling stage of AIS. Defaults to 300.
- **m__vecchia**: An integer defining the maximum number of nearest neighbors utilized in the Vecchia approximation for spatial covariance likelihood calculation. Defaults to 20. Used only when model = "AIS_copula".
- **mcmc__samples**: An integer specifying the total number of MCMC iterations. Defaults to 1000.
- **burn__in**: An integer specifying the number of initial MCMC iterations to discard as burn-in. Defaults to 1000.

Value

A named list containing post-burn-in MCMC samples (length `M`):

- **theta**: A numeric vector of length M containing posterior draws for the coefficient θ for the unobserved predictor ζ.
- **beta**: A numeric matrix of dimension `M x p` containing posterior draws for known covariate regression parameters X (returned if `x` is provided).
- **sigma__y__sq / r**: A numeric vector of length M containing posterior draws for residual observation variance σ_ϵ^2 (Normal) or dispersion size parameter r (Negative Binomial).
- **zeta__sam**: A numeric matrix of dimension `M × n` containing posterior samples of ζ.

Other outputs can exist for custom `sample_params`, `init_params`, and `lik_fun`.

`init_params_normal`

Generates initial starting values for regression coefficients for known predictors (β), the regression coefficient for unobserved ζ (θ), and the Gaussian residual variance (σ_ϵ^2).

Usage

```
init_params_normal(y, x = NULL, zeta_ppd)
```

Arguments

- **y**: A numeric vector of length `n` containing outcome data
- **x**: A `n x p` matrix representing known fixed-effect covariate predictors. Defaults to `NULL`
- **zeta__ppd**: A numeric matrix of dimension `S × n` containing `S` posterior predictive draws of unobserved variable ζ obtained from Stage 1.

Value

A named list containing: * **beta**: Initial covariate coefficients (numeric vector of length p). * **theta**: Initial spatial scale parameter (scalar initialized to `1.0`). * **sigma__y__sq**: Initial residual variance (scalar initialized to `1.0`).

`sample_params_normal`

Executes a conjugate Normal-Inverse-Gamma Gibbs sampling step for standard linear regression parameters and observation noise variance conditional on the current draw of ζ.

Usage

```
sample_params_normal(
  y,
  x = NULL,
  zeta_sam,
  params,
  prior_prec = 0.001,
  a_sigma = 0.001,
  b_sigma = 0.001
)
```

Arguments

- **y**: A numeric vector of length `n` containing outcome data
- **x**: A `n x p` matrix representing known fixed-effect covariate predictors. Defaults to `NULL`
- **zeta_sam**: A numeric vector of length `n` representing the current draws of ζ.
- **params**: A named list containing current parameter values (`beta`, `theta`, `sigma_y_sq`).
- **prior_prec**: Prior precision scalar for the Gaussian prior on regression parameters. Defaults to `0.001`.
- **a_sigma**: Inverse-Gamma shape hyperparameter for σ^2_ϵ. Defaults to `0.001`.
- **b_sigma**: Inverse-Gamma rate hyperparameter for σ^2_ϵ. Defaults to `0.001`.

Value

A named list containing updated parameter values (`beta`, `theta`, `sigma_y_sq`).

`lik_fun_normal`

Evaluates the pointwise Gaussian log-likelihood matrix of continuous outcome vector y across all `S` Stage 1 partial posterior draws of ζ under current parameter values.

Usage

```
lik_fun_normal(y, x = NULL, zeta_ppd, params)
```

Arguments

- **y**: A numeric vector of length `n` containing outcome data
- **x**: A `n x p` matrix representing known fixed-effect covariate predictors. Defaults to `NULL`
- **zeta_ppd**: A numeric `S x n` matrix containing Stage 1 partial posterior draws of ζ.
- **params**: A named list containing current parameter values (`beta`, `theta`, `sigma_y_sq`).

Value

A numeric matrix of dimension `S × n` containing log-likelihood values.

`sample_zeta_iis`

Draws a single vector ζ_{sam} using Independent Importance Sampling (IIS).

Usage

```
sample_zeta_iis(log_lik, zeta_ppd)
```

Arguments

- **log_lik**: A numeric matrix of dimension `S × n` containing log-likelihood values evaluated by `lik_fun`.
- **zeta_ppd**: A numeric `S x n` matrix containing Stage 1 partial posterior draws of ζ.

Value

A numeric vector of length `n` representing the selected draw of ζ from `zeta_ppd`.

sample_zeta_ais

Performs Adjusted Importance Sampling (AIS) without spatial copula or Vecchia re-weighting.

Usage

```
sample_zeta_ais(log_lik, zeta_ppd, R = 300)
```

Arguments

- **log_lik**: A numeric matrix of dimension `S × n` containing log-likelihood values evaluated by `lik_fun`.
- **zeta_ppd**: A numeric `S x n` matrix containing Stage 1 partial posterior draws of ζ.
- **R**: An integer defining the number of discrete candidate draws resampled in the second sampling stage of AIS. Defaults to 300.

Value

A numeric vector of length `n` representing the selected draw of ζ from `zeta_ppd`.

transform_to_z

Transforms candidate ζ draws from their original scale into standard Gaussian latent variables $Z\ N(0, 1)$. Uses a continuous piecewise-linear Probability Integral Transform (PIT) via empirical CDF interpolation.

Usage

```
transform_to_z(zeta_mat, sorted_zeta_ppd)
```

Arguments

- **zeta_mat**: A numeric matrix of dimension `R x n` containing `R` candidate draws across `n` spatial locations.
- **sorted_zeta_ppd**: A numeric matrix of dimension `S x n` containing sorted Stage 1 partial posterior draws for each location.

Value

A numeric matrix of dimension `R x n` containing transformed standard normal Z-scores.

sample_zeta_ais_copula_vecchia

Executes two-layer Adjusted Importance Sampling (AIS). Layer 1 samples candidate draws based on pointwise likelihoods; Layer 2 adjusts proposal weights using a spatial Gaussian copula density evaluated via the Vecchia approximation.

Usage

```
sample_zeta_ais_copula_vecchia(
  log_lik,
  zeta_ppd,
  sorted_zeta_ppd,
  locs,
  covparms,
  NNarray,
  R = 300,
  m = 20
)
```

Arguments

- **log_lik**: A numeric matrix of dimension `S × n` containing pointwise log-likelihood values evaluated by `lik_fun`.
- **zeta_ppd**: A numeric matrix of dimension `S × n` containing Stage 1 partial posterior draws.
- **sorted_zeta_ppd**: A numeric matrix of dimension `S × n` containing column-sorted Stage 1 draws.
- **locs**: A numeric matrix of dimension `n × 2` specifying spatial coordinates.
- **covparms**: A numeric vector specifying spatial covariance parameters (variance, range, smoothness, nugget).
- **NNarray**: An integer matrix of dimension `n × m` containing pre-computed ordered nearest-neighbor indices generated by `GpGp::find_ordered_nn`.
- **R**: An integer specifying the number of draws taken from the proposal distribution for AIS. Defaults to `300`.
- **m**: An integer specifying the maximum number of nearest neighbors in the Vecchia approximation. Defaults to `20`.

Value

A numeric vector of length `n` representing the selected ζ_{sam} for the current MCMC iteration.

`init_params_negbin`

Generates initial values for regression parameters (β, θ) and the Negative Binomial size parameter r.

Usage

```
init_params_negbin(y, x = NULL, zeta_ppd)
```

Arguments

- **y**: A numeric vector of length `n` containing outcome count data
- **x**: A `n x p` matrix containing known fixed covariates. Defaults to `NULL`
- **zeta_ppd**: A numeric matrix of dimension `S × n` containing Stage 1 partial posterior draws.

Value

A named list containing:

- **beta**: Initial covariate coefficients (numeric vector of length p).
- **theta**: Initial spatial scale parameter (scalar initialized to 1.0).
- **r**: Initial dispersion/size parameter (scalar initialized to 2.0).

lik_fun_negbin

Evaluates the pointwise log-likelihood matrix of Negative Binomial outcomes across all partial posterior draws of ζ and current parameter values.

Usage

```
lik_fun_negbin(y, x = NULL, zeta_ppd, params)
```

Arguments

- **y**: A numeric vector of length `n` containing outcome count data
- **x**: A `n x p` matrix containing known fixed covariates. Defaults to `NULL`.
- **zeta_ppd**: A numeric matrix of dimension `S × n` containing Stage 1 partial posterior draws.
- **params**: A named list containing current parameter values (`beta`, `theta`, `r`).

Value

A numeric matrix of dimension `S x n` containing log-likelihood values.

slice_sample_r

Updates the Negative Binomial dispersion parameter r using a 1D univariate slice sampler under a Cauchy-like prior $p(r) \propto (1 + r^2 e_0)^{-1}$

Usage

```
slice_sample_r(r_curr, y, eta, e0 = 0.01, a_r = 0.001, b_r = 100)
```

Arguments

- **r_curr**: Numeric scalar representing the current state of `r`.
- **y**: A numeric vector of length `n` containing outcome count data
- **eta**: A numeric vector of length `n` containing current linear predictor values $\eta = X\beta + \theta\zeta$.
- **e0**: Prior hyperparameter scale factor. Defaults to `0.01`.
- **a_r**: Numeric lower bound for r. Defaults to `0.001`.
- **b_r**: Numeric upper bound for r. Defaults to `100`.

Value

A numeric scalar representing the updated sample of r.

sample_params_negbin

Implements a Gibbs sampler step for Negative Binomial model parameters. Uses Polya-Gamma latent auxiliary variables ($\omega_i \sim PG$) to sample regression parameters (β, θ) conditionally from Gaussian distributions, followed by a slice sampling step for dispersion parameter `r`.

Usage

```
sample_params_negbin(
  y,
  x = NULL,
```

```
  zeta_sam,
  params,
  prior_prec = 0.001,
  e0 = 0.01,
  a_r = 0.001,
  b_r = 100
)
```

Arguments

- **y**: A numeric vector of length `n` containing outcome count data
- **x**: A `n x p` matrix containing known fixed covariates. Defaults to `NULL`.
- **zeta_sam**: A numeric vector of length `n` representing the current sampled draws of ζ.
- **params**: A named list containing current parameter values (`beta`, `theta`, `r`).
- **prior_prec**: Prior precision scalar for Gaussian regression coefficients. Defaults to `0.001`.
- **e0**: Prior hyperparameter scale factor. Defaults to `0.01`.
- **a_r**: Numeric lower bound for r. Defaults to `0.001`.
- **b_r**: Numeric upper bound for r. Defaults to `100`.

Value

A named list containing updated parameter values (`beta`, `theta`, `r`).

`setup_copula_cache`

Pre-computes and caches persistent structures required for Layer 2 spatial Gaussian copula evaluations in AIS_copula. This includes sorting Stage 1 draws for continuous empirical CDF mapping, constructing Vecchia nearest-neighbor arrays, and fitting spatial covariance parameters with standardized Z-space variance and an explicit nugget parameter floor to prevent spatial over-smoothing.

Usage

```
setup_copula_cache(zeta_ppd, locs, m_vecchia = 20, nugget_prop = 0.10)
```

Arguments

- **zeta_ppd**: A numeric matrix of dimension `S × n` containing Stage 1 partial posterior draws.
- **locs**: A numeric matrix of dimension `n × 2` specifying spatial coordinates.
- **m_vecchia**: An integer defining the maximum number of nearest neighbors utilized in the Vecchia approximation for spatial covariance likelihood calculation. Defaults to 20.
- **nugget_prop**: A numeric scalar in `[0,1)` defining the proportion of total Z-space marginal variance allocated to the nugget floor. Defaults to `0.10` (10% nugget). Used to prevent slope attenuation.

Value

A named list containing:

- **sorted_zeta_ppd**: A numeric matrix of dimension `S x n` containing sorted Stage 1 partial posterior draws for each location used by transform_to_z for fast interpolated Probability Integral Transforms (PIT).
- **NNarray**: An integer matrix of dimension `n × m` containing ordered nearest-neighbor indices generated by `GpGp::find_ordered_nn`.
- **covparms**: A numeric vector of specifying spatial covariance parameters (variance, range, nugget).